

Machine learning project proposal - setMaker

Problem statement:

For a given tune, predict a number of other tunes that are likely to fit well in a set together based on features and metadata extracted from the tune as well as historical set data.

Horizon and success criteria:

The model should be able to generate 10 tune predictions within a reasonable time frame of 1 second of the user inputting either a known or previously unknown tune. If the tune is featured in the dataset, the models recommendations should feature at least 40% of tunes that have appeared after the input tune in the historical set data, also formulated as $\text{Recall}@10 \geq 0.40$. The model should also try outperform a simple baseline, where the 10 predictions are based on the frequency of tunes appearing in the same set after the input tune in the historical data in it's training data.

Background:

The Irish session culture is one that has spread all around the globe, both due to the mass-emigration of Irishmen throughout history, and the popularity with local musicians and audiences. As an effect, the musical traditions of a session has become almost standardised to the point where you will find that the format is almost identical, whether you are in Zürich, Wellington or Dublin. In this tradition, you are expected to play tunes (singular melodies that can be thought of a bit like songs) together in sets; that is playing multiple tunes after one another. The art of crafting sets is not trivial, and is something many musicians put a great deal of pride and effort into, it can be seen a bit like the craft of making good playlists, or dj's mixing songs together. The site: thesession.org is a database over tunes where users can upload tunes, comment, and relevantly for us, upload sets. The site has become very well-known and is used as a global archive of tunes, hosting over 55'000 as of writing, with new tunes being added every day.

Idea and motivation:

The craftsmanship of making sets is a complicated one with two sides to it, there are some very empirical measurement that determines if two tunes fit well together, like their key signature, type, meter, tonal density, and much more, but a set is also judged by it's creativity, and often, it's ability to hit a spot, where the tunes in question is not too similar but not too different.

This is why it is an interesting ML problem, since making an algorithm to find tunes that are the most similar would be easy, but with historical data of sets to work with, we could potentially train a model to hit the sweet spot of making coherent, yet interesting sets.

I have a personal interest in this topic as a session musician myself. I think this data could be interesting to work with, and while my other ideas had problems with originality, this project certainly stands out among projects that has been done before.

Data source:

The entire database of tunes, sets and much more from thesession.org can be found on github, where it is updated once a week. Since our data by no means is time-sensitive, and our input is provided by the user, we do not need the data as soon as it is uploaded to thesession.org but can wait till the github-repo is updated. As of writing the dataset contains 55'000 tunes, 150'000 sets and 202'000 recorded tracks. There are uploaded around 6 tunes and 200 sets a day.

Features:

The data for a tune includes metadata, and the tune itself notated as "ABC's". This is a music-notation format that gives us the raw melody, kind of like sheet-music. Since the metadata is too limited to make good predictions on, our system will need to interpret the "ABC's" data as well.

This will be done by a separate parser that will interpret the "ABC's" and return features such as: length, pitch range, rhythm features, repeating patterns and more. Exactly which features that should be derived from this process should be tested and experimented with during development.

Labels:

The labels will come from both the set-data and the recordings-data. The majority of Irish music recordings feature sets, so we are also able to derive set-data from this. Sets can feature more than two tunes, but this model will only be trained to predict one tune to play after the input tune. This means the label-data will be carved up, so the set "A→B→C" will be made "A→B" and "B→C".

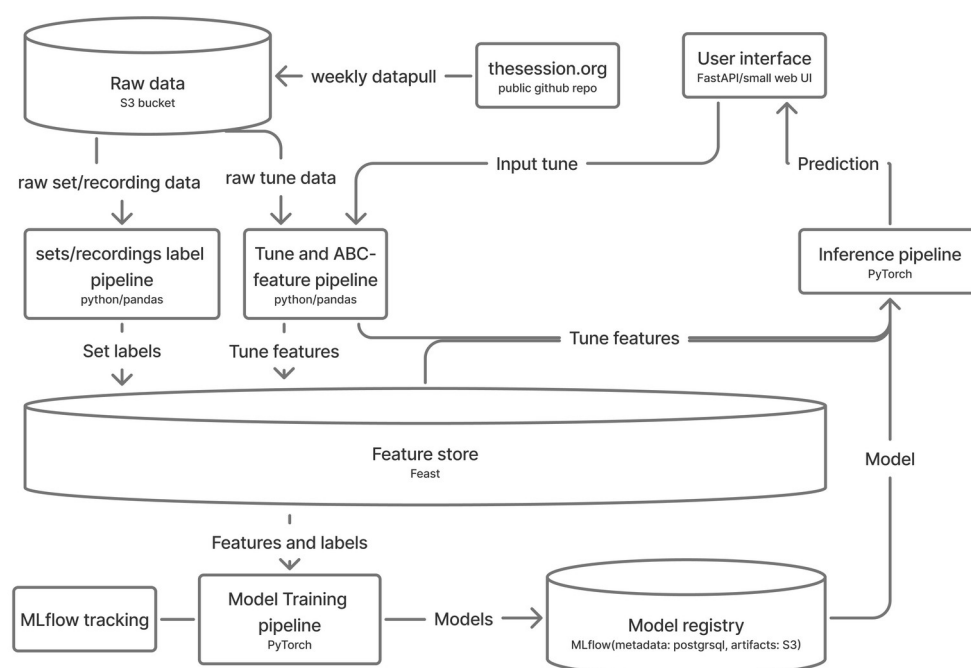
Rare positive:

Since the majority class is negative, we can evaluate the model based on Precision@10 or similar, this meaning that since our model will give 10 predictions, we can rank how many positives; meaning tunes that were played together in the set data, was predicted in the top 10 predictions. If this is too little, we could even increase the number of predictions per input.

Leakage and training:

Since we have a timestamp for all data, it would make sense to divide the training/testing data based on time of upload. This way we avoid leakage. Some issues might rise from having duplicate label data, due to carving up sets, fx we could encounter "A→B→C" in the training data, and then "A→B→D" in the test data. Both of these would result in having duplicate "A→B" sets

FTI diagram:



Tech stack:

Feature store	Parquet in S3 bucket	S3 is chosen since we can version by build date, this gives us precise point in time data.
Experiment tracking	ML flow	MLflow can be used in both tracking and model registry.
Orchestration	Prefect	The orchestration in this project is very lite, and Prefect seems easy to set up.
Serving	FastAPI	A simple API set up in a docker container is sufficient.
Testing	Github actions	Is integrated with github, and is easy to set up.

It could down the line be cool to build IaC for the project intergraded with a cloud provider, so you ideally could build the entire system in the cloud with a single command, but this is definitely a stretch depending on time constraints.

Relevant links:

Public github repo -> <https://github.com/Christian-Skovgaard/setMaker>

Thesession.org -> www.thesession.org

The session data github repo -> <https://github.com/adactio/TheSession-data>

Youtube video of a session if you are curious and unfamiliar -> <https://youtu.be/DlrGbTIZQ0>